# Vitamin A Deficiency (VAD)

**Level 4 Ophthalmology lecture, MB,ChB, November 2015**

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# What is vitamin A?

It is a **fat soluble "essential vitamin"** whose functions include:

- Vision (visual pigments)
- Gene expression and cell differentiation
- Growth
- Immunity
- Antioxidant

Lack of Vitamin A leads to Vitamin A Deficiency Disorders (**VADD**)

# Sources of vitamin A

- Occurs as retinol in animal sources:
  - Breast milk
  - Egg yolk
  - Dairy products
  - Fish liver
- Pro-vitamin-A precursors in plants (carotenoids such as β-carotene):
  - Yellow, orange and red fruits (orange, mango, papaya)
  - Yellow and orange vegetables (carrots, squash)
  - Dark green leafy vegetables (spinach, sukuma wiki)
  - Red palm oil (West Africa)

# CAUSES OF VAD

- Dietary deficiency
- Mal-absorption
  - GIT disease
  - Intestinal worms
- Uptake deficiency
  - Liver disease
- Protein calorie malnutrition
- Increased demand
  - Disease → fever, loss of epithelium e.g. measles

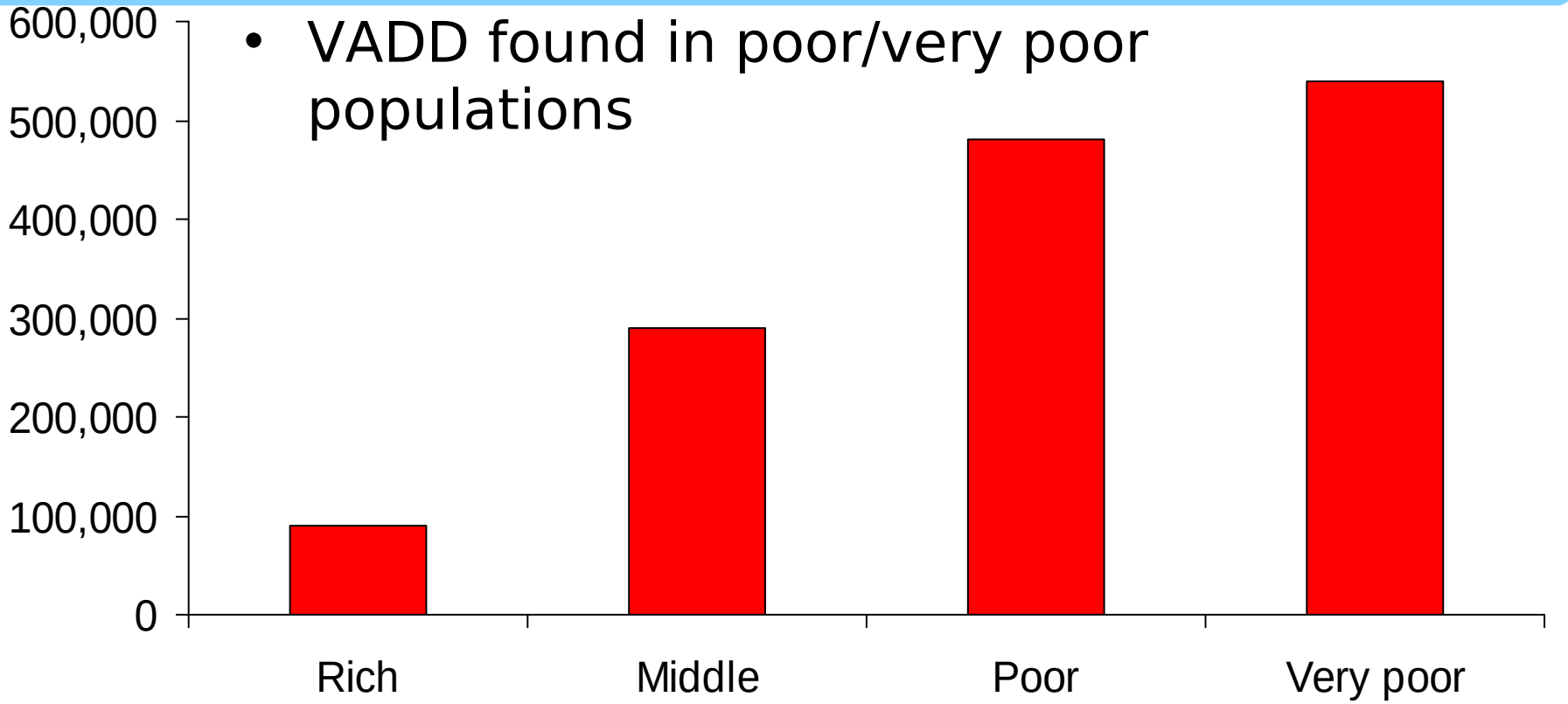
# Mortality, blindness and VAD

- There is approximately 1.4 million blind children in the world of whom 350,000 (**25%**) are blind due to VAD
- Generally, major causes of childhood blindness are major killers
- Under-5 mortality rate is often used as proxy indicator for the prevalence of childhood blindness
- Even sub-clinical VAD has been associated with high mortality

- **VAD is rare in adults.** Those at risk include:
  - Pregnant/lactating mothers
  - Prisoners
  - Refugees/internally displaced persons

# Where do the world's blind children live?

- Total = 1.4 million blind children
- VADD found in poor/very poor populations



# Vitamin A Metabolism

- Carotenes are changed to **retinol** at gut wall
- Retinol is **best absorbed with fried foods** since it is fat soluble (the poor often lack cooking fat/oil)
- Liver stores Vitamin A and synthesize Retinal Binding Protein (RBP)
- Retinol is bound to RBP and in complex with transthyretin, it is transported to the eye and epithelia via blood

- Normal serum levels = 1.0 to 1.4µmol/L

- Low = 0.7 to <1.0µmol/L and deficient = <0.7µmol/L

# Assessment of individuals for Vitamin A deficiency

- **Clinical** VAD: clinical signs when serum levels are <0.7µmol./L
- **Sub-clinical** VAD is confirmed using laboratory tests:

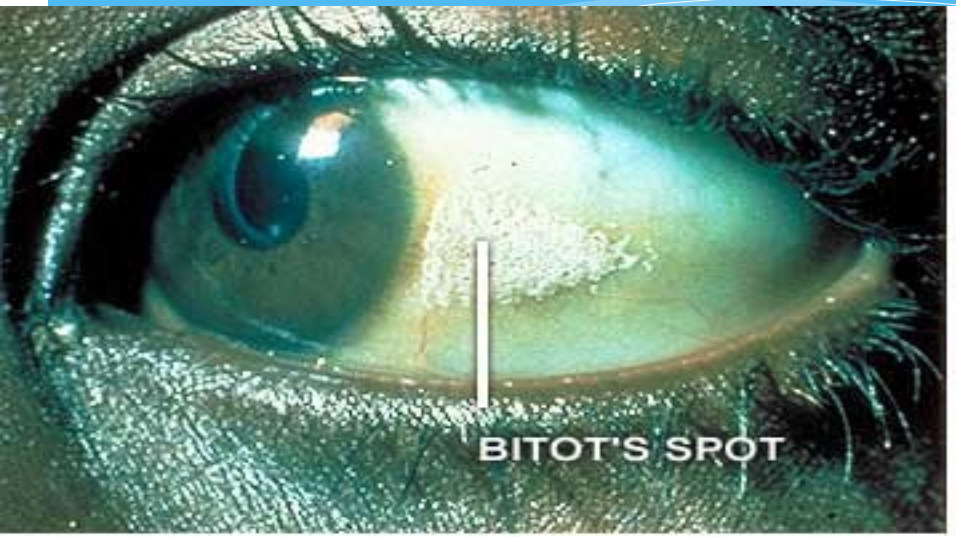
  - Histology = CIC (conjunctival impression cytology)

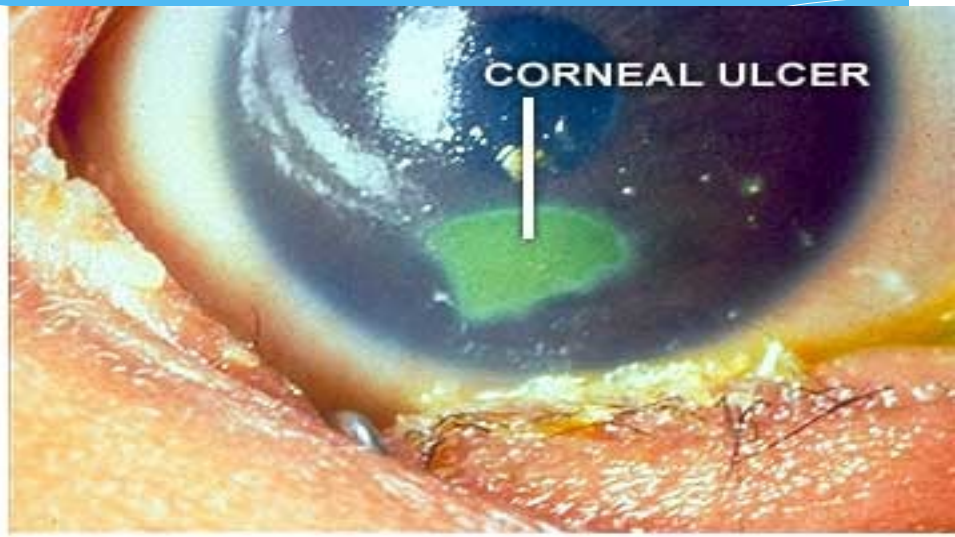
  - Biochemical (serum retinol) = relative dose response (RDR) and modified RDR
- **Functional** impairment such as night blindness rate (History from mothers) can also be used as an indicator for VAD
- Others: local name for XN (not reliable)

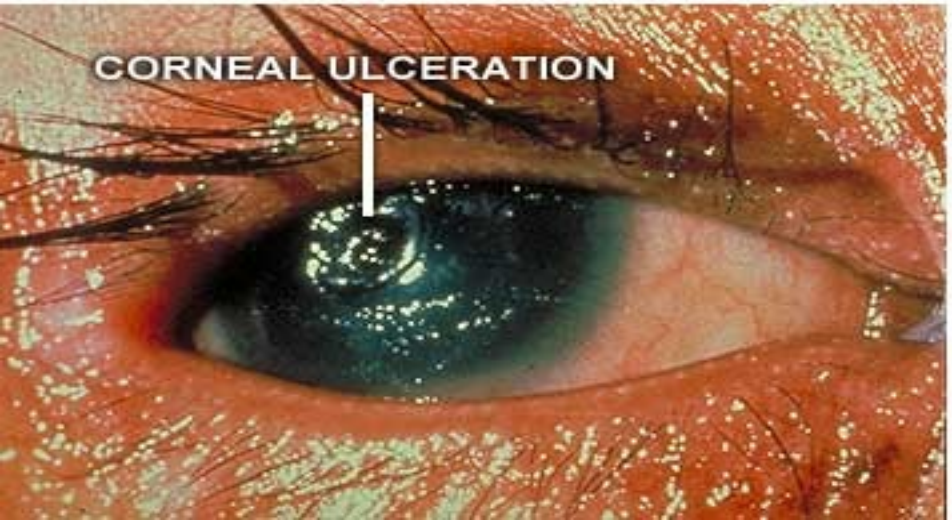
# VAD SIGNS

- Xeropthalmia (‘dry eye’)
- Corneal ulceration
- Night blindness
- Lowered host resistance (risk of systemic infection)

# Ocular signs of VAD

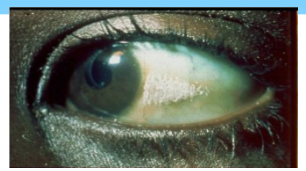


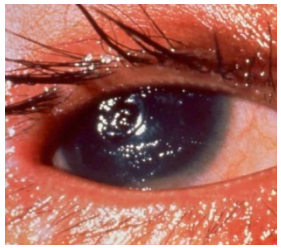



# WHO grading of VAD (Xerophthalmia)

- XN = Night blindness
- X1A = Conjunctival Xerosis
- X1B = Bitot spot
- X2 = Corneal Xerosis
- X3A = Corneal ulcer/keratomalacia <1/3
- X3B = Corneal ulcer/keratomalacia >1/3
- XS = Xerophthamia corneal scar
- XF = Xerophthamia fundus
- X5 = Corneal scar





# Peak age groups of VAD signs

**Grade of xerophthalmia Peak age group affected**

- XN = Night blindness 2-6 years and women
- X1A = Conjunctival Xerosis 3-6 years
- X1B = Bitot spot 1-4 years
- X2 = Corneal Xerosis 1-4 years
- X3A = Corneal ulcer/keratomalacia <1/3 1-4 years
- X3B = Corneal ulcer/keratomalacia >1/3 1-4 years
- XS = Xerophthamia Corneal scars 1-4 years and older
- XF = Xerophthamia fundus Adults

# X1A

- Loss of goblet cells (mucin) and keratinisation
- Squamous metaplasia
- Improves after 2-4 days of treatment

# X1B

- Foamy spot (bitot)
- usually temporal
- Keratinisation and corynebacteria xerosis
- Takes long to disappear after treatment

# Corneal Xerosis/Keratomalacia

- X2: dry cornea due to keratinisation (reversible with treatment)
- X3A and X3B leave scar after healing (potentially blinding)
- Keratomalacia: cornea melts (stromal necrosis). It can occur very rapidly without warning signs and eye may rapture
- Children 1-5 yrs old (preschool age group) are affected most

# Assess VAD status in the community

## Minimum prevalence criteria

(for VAD to be declared a public health problem)

| Clinical indicator | Minimum prevalence |
|---|---|
| XN | 1% |
| XB | 0.5% |
| X2 and X3(A,B) | 0.01% |
| XS | 0.05% |
| Serum retinol (<0.35µmol/L) | 5% |

# Risk factors of VAD

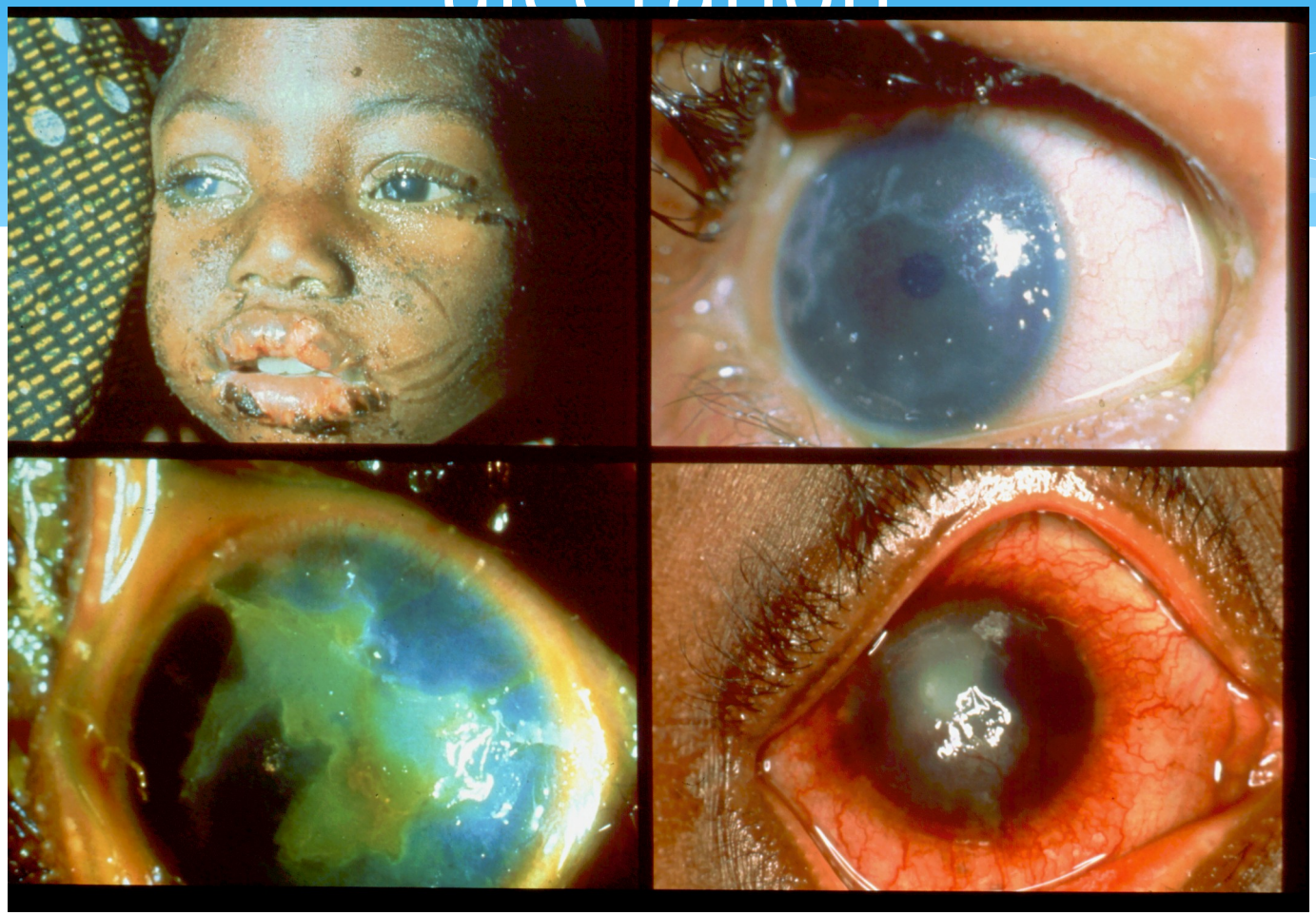
- **Poverty**: no land, water, health facilities and infrastructure
- **Age**: children have high demand, poor liver stores and frequent illnesses
- **Malnutrition/early** weaning (breast feeding)/food taboos
- **Diseases**: measles, diarrhea, febrile illnesses
- Traditional medical practices may complicate/delay treatment
- Maternal education
- Wars/conflicts (internally displaced person/refugees)

# Measles and VAD corneal ulceration

Measles results in:

- **High demand** for Vitamin A: child, fever, loss of epithelial cell
- **Poor intake** of vitamin A: sole mouth, nausea and vomiting
- **Poor absorption** and increased loss of vitamin A and amino acids: sore gut vomiting and diarrhea
- Co-existing **malnutrition** make it worse (RBP/corneal healing)
- **Viral invasion** of the cornea: measles/HSV and dehydration: dry exposed cornea
- Clinical signs may be worsened by **TEM**

# Measles and Herpes Simplex Viral corneal ulceration



# Malnutrition

- Interferes with vitamin A absorption and transport
- PEM (Kwashiokor in particular) destabilizes the structural proteins including corneal collagen
- Lower immunity make the cornea vulnerable. Measles rash (and immune reaction) may not be noted hence misdiagnosed
- Malnutrition worsen measles

# Few malnourished sick children with VAD will survive

The PHOTO in the mail is the "Pulitzer prize" winning photo taken in 1994 during the Sudan famine. The picture depicts a famine stricken child crawling towards an United Nations food camp, located a kilometer away.

>The vulture is waiting for the child to die so that it can eat it. This picture shocked the whole world. No one knows what happened to the child, including the photographer Kevin Carter who left the place as soon as the photograph was taken.

>Three months later he committed suicide due to depression.

# Harmful Traditional Eye Medicines (HTEM)

- **Not all TEM are harmful** to the eye
- However, some including plant extracts/sap, powders, ash and urine may:
  - Result in corneal infection if they are **not be sterile**
  - Be corrosive because of high **pH** and **concentration** (painful)

# TREATMENT OF XEROPHTHALMIA

* **Children over one year old.**
  * Immediately on diagnosis (Day 1): **200,000 IU vitamin A orally**
  * The following day (Day 2) : 200,000 IU vitamin A orally
  * Four weeks later (Week 4): 200,000 IU vitamin A orally
  * If there is vomiting, an intramuscular injection of 100,000 IU of water soluble vitamin A (not an oil-based preparation) may be used instead of the first oral dose.
* **Children under one year old or < 8 kg** → Use half the doses of the regimen given above.

# Prevention of disease

- **Primary prevention** = make health people remain healthy (Dietary intake)
- **Secondary prevention** = treat people with underlying disease to prevent disability
- **Tertiary prevention** = treat those with disability (e.g. blindness) to make them see again (e.g. cataract, cornel graft) or rehabilitation

# Control of VAD

**Short term measures** (intermittent high dose vitamin A supplementation)

- **Treat cases**: 200,000iu stat, on day 2 and then after 4 weeks (half the dose for children <1yr or <8Kg)
- Treat those at risk (**prophylaxis**): sick/malnourished children, pregnant/breast feeding women and prisoners/remand

# Control of VAD

**Mid term measures**: immunization and food fortification and supplementation programmes

- Measles immunization
- Fortify foods
- Control diarrhoeal diseases

# Control of VAD

**Long term measures:** address underlying causes/risk factors

- Improve nutrition of mothers and children
- Kitchen gardens
- Nutrition health education
- NB. Macro and micro economic policies formulations

# Prevention of VAD

# Remember children

- Their vitamin A liver stores are relatively low at birth
- Breast milk supply adequate retinol for the first 6 months
- Their daily requirements are high because are growing
- They have frequent infections and diarrhoea after weaning
- They may contract measles after the age of 6 months
- They are totally dependant on you (adults)

# THANK YOU